LipScribe: Teaching a Computer to Read Lips and Listen at the Same Time

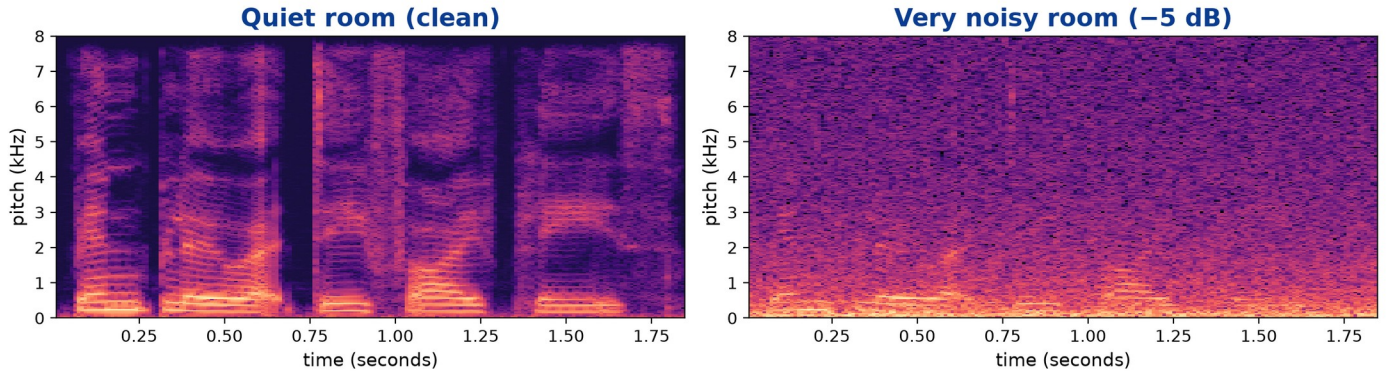
Progress so far (Weeks 1–4) · Computer Vision and Deep Learning · Group 4

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The problem: captions break when a room gets loud



A ‘sound picture’ of the same sentence. The bright stripes are the voice; on the right, noise almost hides them.

Who this helps: people who are deaf or hard of hearing, and anyone captioning a lecture, market, taxi park or video call where the audio is poor.

1 Sound gets buried

Noise drowns out the voice, so a listen-only system starts guessing.

2 Lips don’t change

Noise can’t touch what the camera sees. The mouth moves the same way.

3 Our idea: use both

Listen and lip-read together, and lean on the lips when it gets loud.

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Where we are, in four numbers

1.0%

words wrong in a quiet room

about 1 word in 100

0.7%

with our grammar checker on

fewer than 1 word in 100

−43%

fewer errors in the noisiest room

5.3% vs 9.5% for listen-only

1.9%

with the camera switched off

still as good as listen-only

In short: all planned models are built and tested. Our final design, Model P, makes the fewest mistakes overall, cuts errors in noise, and keeps working if the camera or microphone fails.

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Four words you'll hear in this talk

M Model

A program that learns a skill from examples instead of following hand-written rules. We show it thousands of videos with their correct sentences, and it slowly learns the pattern.

% Word error rate (WER)

Out of every 100 words spoken, how many the model gets wrong. 2% means 2 mistakes per 100 words. Lower is better.

dB Noise level (dB)

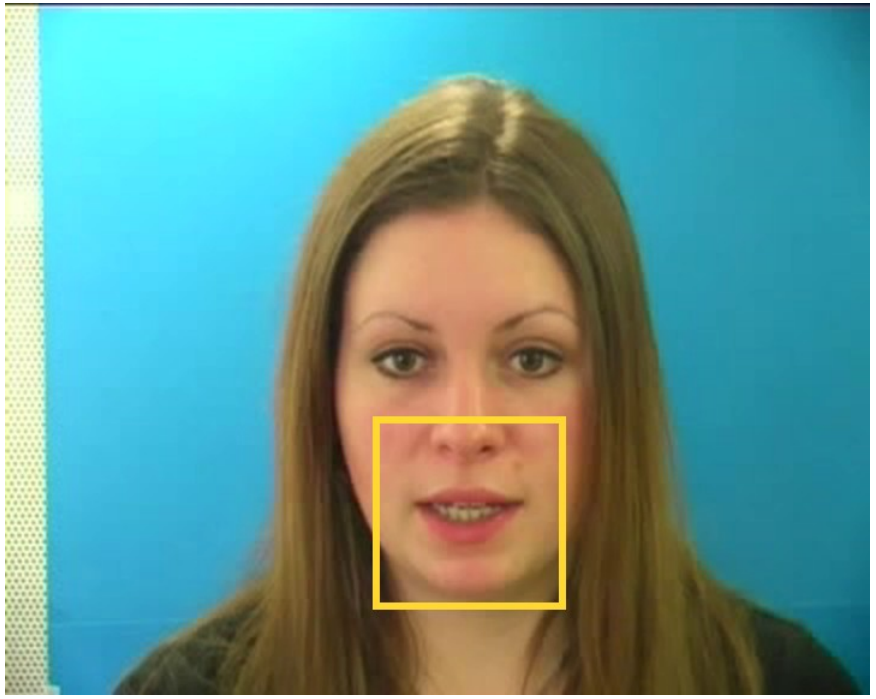
How loud the background is compared to the voice. "Quiet" has no noise; at −5 dB the noise is about 3× stronger than the speaker.

? Unseen speakers

We test only on people the model never saw while learning, so the score shows how it would do on a brand-new person.

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Our data: 33 people, 32,890 short video clips



A real GRID clip. The yellow box is the part we keep.

Every clip is a 3-second, six-word sentence, for example:

“bin blue at d five please”

26
people to learn from

3
people to tune on

4
new people to test on

Quality checks first. An automatic audit of all 34,000 files found one speaker with no video and 110 broken clips. We removed them and kept 96.7% of the data.

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What the computer actually looks at



Six moments from the sentence “bin blue at d five please”: the lip shape changes with each sound.

1 Find the face

A face-finding tool marks 40 points around the lips in every frame.

2 Cut out the mouth

Crop a steady square around the lips, grey, 96×96 pixels.

3 Keep the timing

Never drop a frame: a missed frame is filled in from its neighbours.

4 Line up the sound

Shift the audio so each sound lands on the right lip movement.

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How LipScribe turns a video into a sentence

Lip video

75 mouth pictures

Sound

voice recording

Eye part

learns lip shapes

Ear part

learns sounds

Combiner

merges what the eyes and ears found, moment by moment

Reader + grammar check

spells out letters, then snaps to the nearest valid sentence

“bin blue at d five please”

The key design question is the combiner: how should the system mix the two senses? We tried a simple version first (baselines), then built a smarter one (Model P).

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Progress against our plan

M1	M2	M3–M4	M5	M6–M7	M8–M9
Set up computers and check the data	Prepare all 32,890 clips	Build the three simple models	Build the smart combiner (Model P)	Fine-tune and test thoroughly	Live caption app and demo
Done · 29 Oct	Done · 3 Nov	Done · 4 Nov	Done · 5 Nov	In progress · due 22 Nov	Started · due 29 Nov

Team of six, one shared machine: every model was trained on our Alienware workstation (RTX 4080 SUPER graphics card), and every result was re-checked there on 12 November.

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Step 1: three simple models to beat

B1 Ears only

Listens to the sound.

2.0% wrong in a quiet room

9.5% wrong in the noisiest room

Great when it's quiet, but errors grow almost 5× in noise.

B2 Eyes only

Reads the lips, no sound at all.

23% wrong, in every room

Noise has zero effect, but some sounds look the same on the lips (b, p, m).

B3 Eyes + ears, glued

Sticks both together in one step.

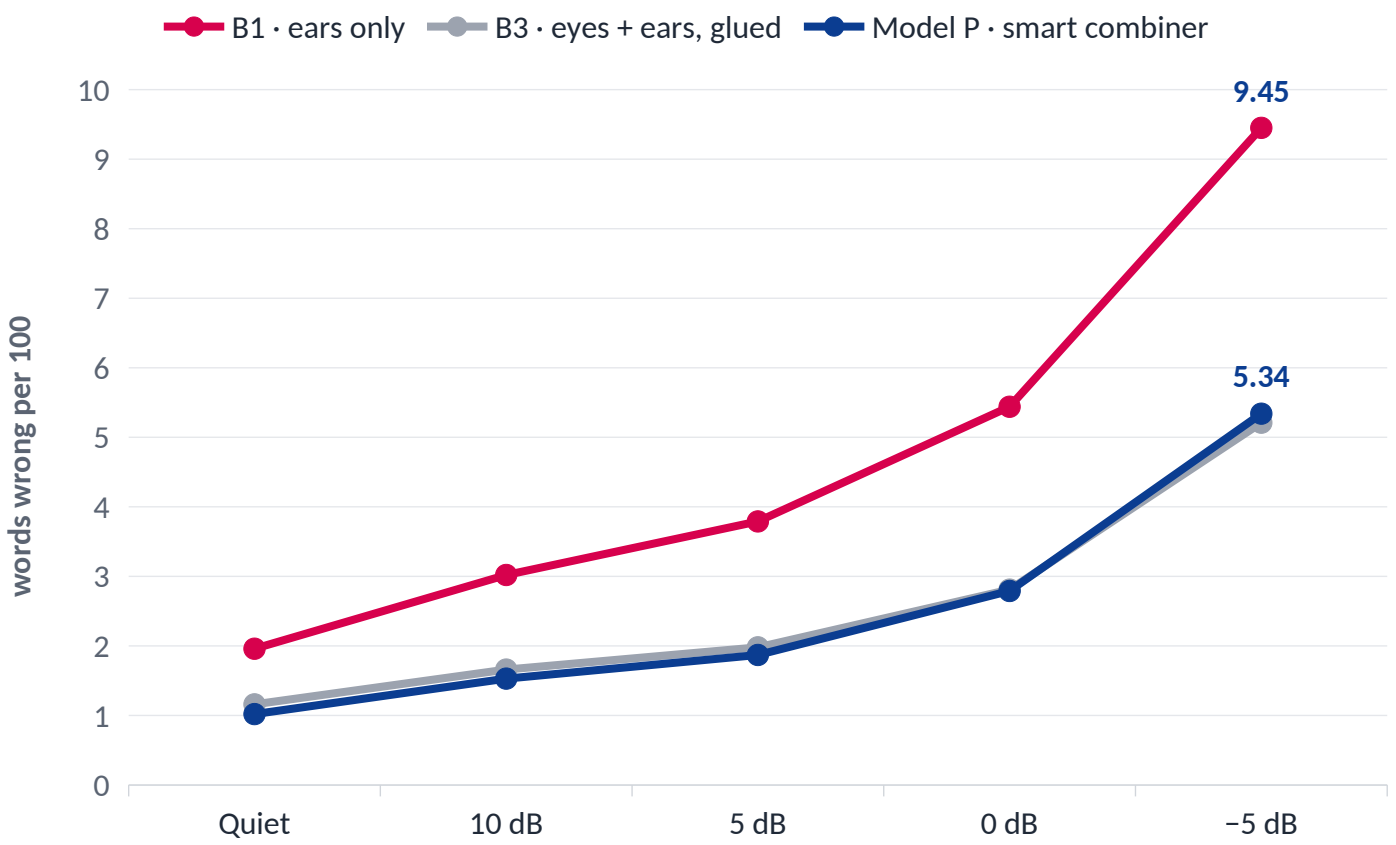
1.2% wrong in a quiet room

5.2% wrong in the noisiest room

Better than B1 everywhere, but it falls apart if the camera or mic stops.

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Using both senses beats listening alone in every room



Quiet room

Model P: 1.0 wrong per 100 words, the best of all our models (ears only: 2.0).

Noisiest room

43% fewer errors than ears only (5.3 vs 9.5). Glued B3 is level here (5.2).

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Why we upgraded the ‘eyes’ to ResNet-18

Old eyes: simple 3D-CNN

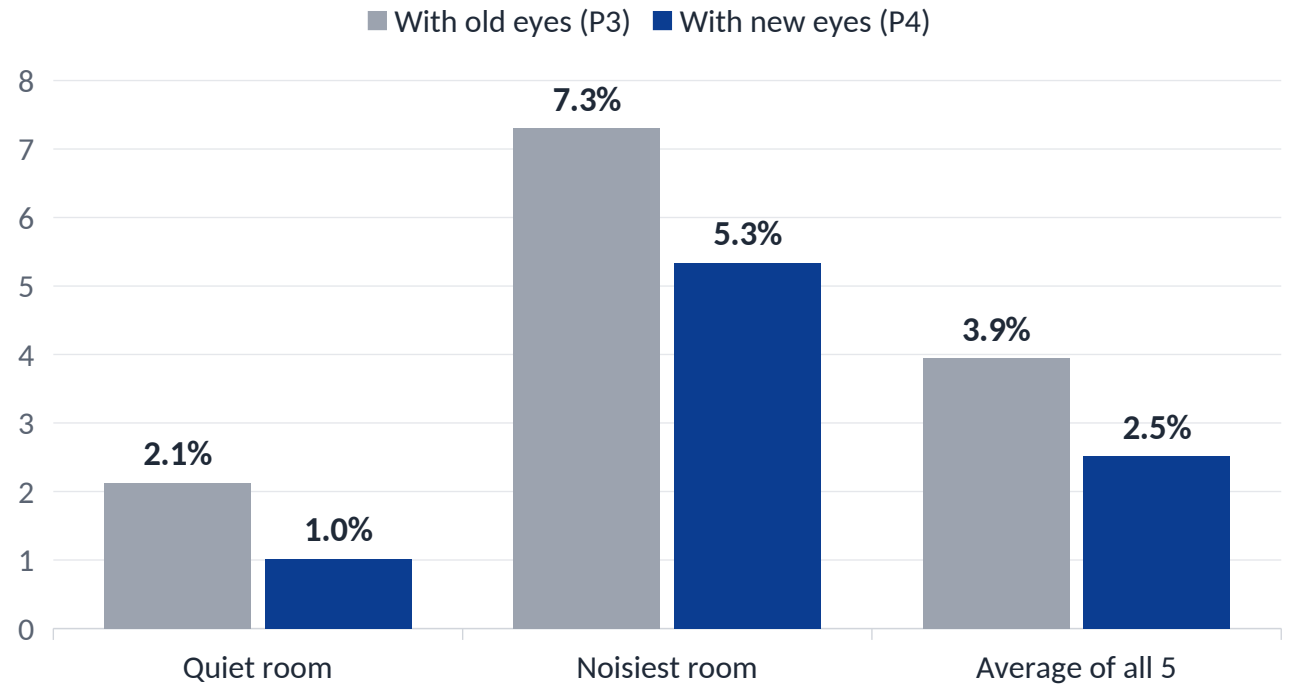
Only 3 layers deep.

Tied to exact pixel positions, so a new face shape or lighting threw it off: 13% to 34% wrong depending on the person.

New eyes: ResNet-18

17 layers with ‘shortcuts’ that make deep learning stable.

Evens out lighting per clip and ignores small shifts of the crop.



Result: half the mistakes in a quiet room (2.1% → 1.0%), and the combiner trusts the lips nearly twice as much in noise.

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Step 2: Model P, a smarter way to combine the senses

1

The senses check each other

Ears can glance at nearby lip movements to fill in a word lost in noise; eyes can use the sound to tell look-alike lip shapes apart (b, p, m).

Technical name: cross-modal attention

2

A trust dial for every moment

25 times a second, the model sets a dial between ‘trust the ears’ and ‘trust the lips’. Loud room: turn towards the lips.

Technical name: gated fusion

3

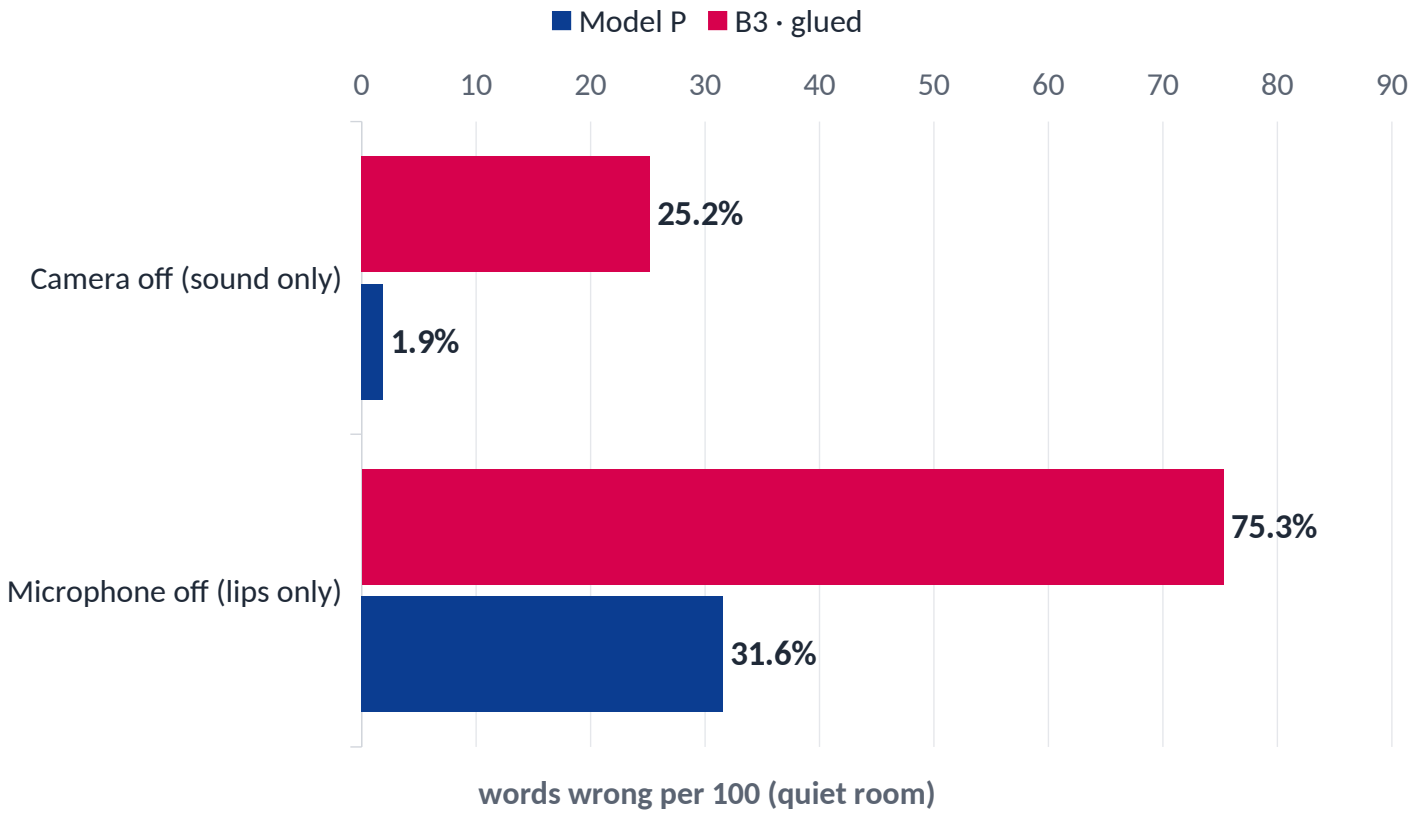
A stand-in when a sense is missing

If the camera loses the face or the mic cuts out, a learned ‘nothing here’ signal fills the gap so the model knows to rely on the other sense.

Technical name: missing-modality embedding

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If the camera or mic fails, Model P keeps going



Camera off

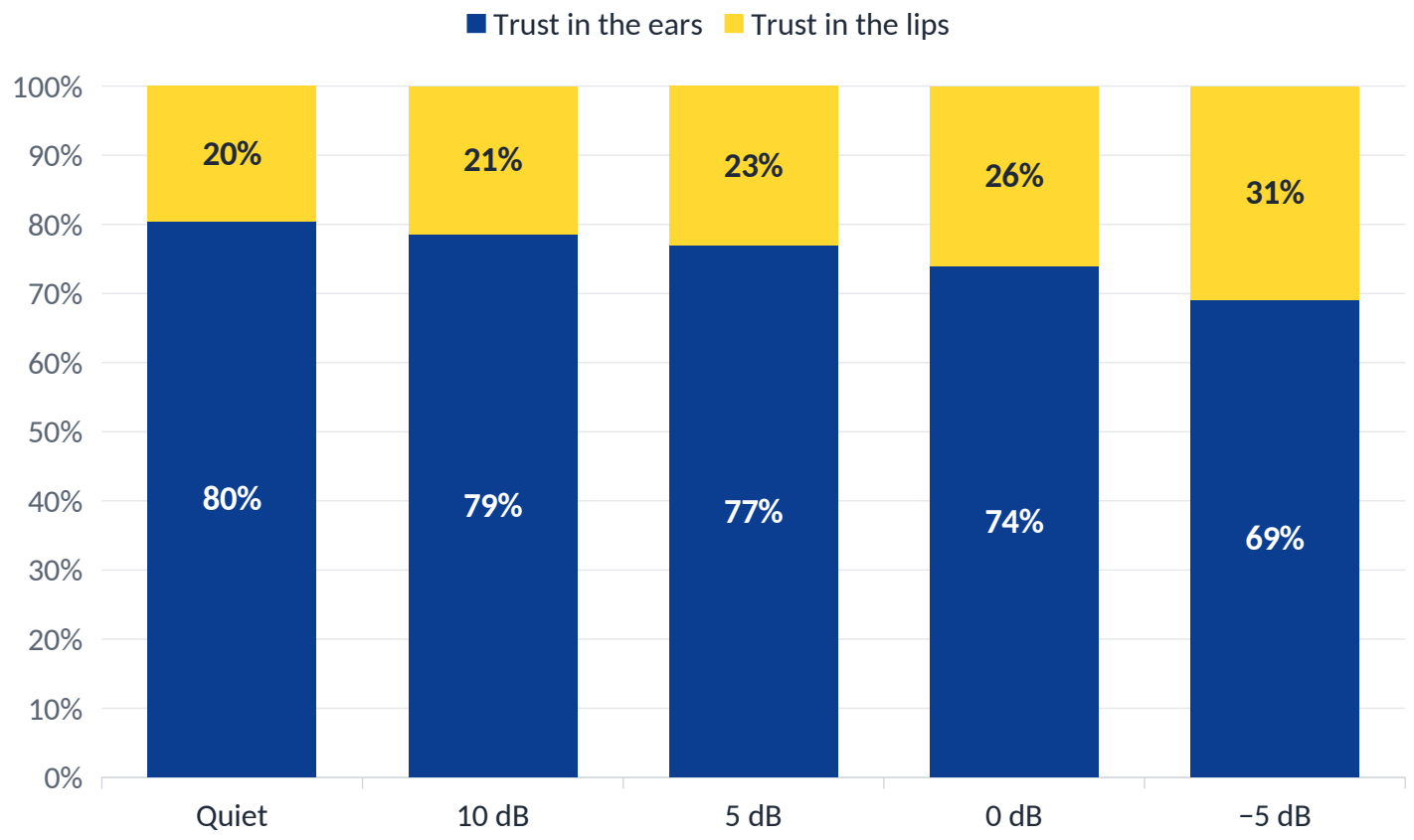
Model P: 1.9% wrong, as good as the ears-only model (2.0%). Glued B3: 25%.

Microphone off

Model P: 32% wrong, as good as lips-only. Glued B3: 75%, nearly useless.

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As the room gets louder, Model P trusts the lips more



20% → 31%

share of trust given to the lips, quiet room → noisiest room

Why it matters

The model learned by itself to lean on the lips when the sound gets bad: exactly the behaviour we designed it for.

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Problems we hit, and how we fixed them

1 Sound ran ahead of the lips

The audio files had their silence cut off, so speech started up to 1.5 s early. We re-aligned every clip using its timing file.

2 Missing and broken files

One speaker had no video and 110 clips wouldn't open. An automatic audit found and removed them.

3 The model 'cheated'

In a practice run all 8 sentences began "bin blue at f", so it guessed the same answer for all. Mixed examples forced it to really read lips.

4 Eyes didn't work on new people

The old eye part ranged from 13% to 34% wrong depending on the person. Fixed by switching to ResNet-18.

5 Gluing made quiet rooms worse

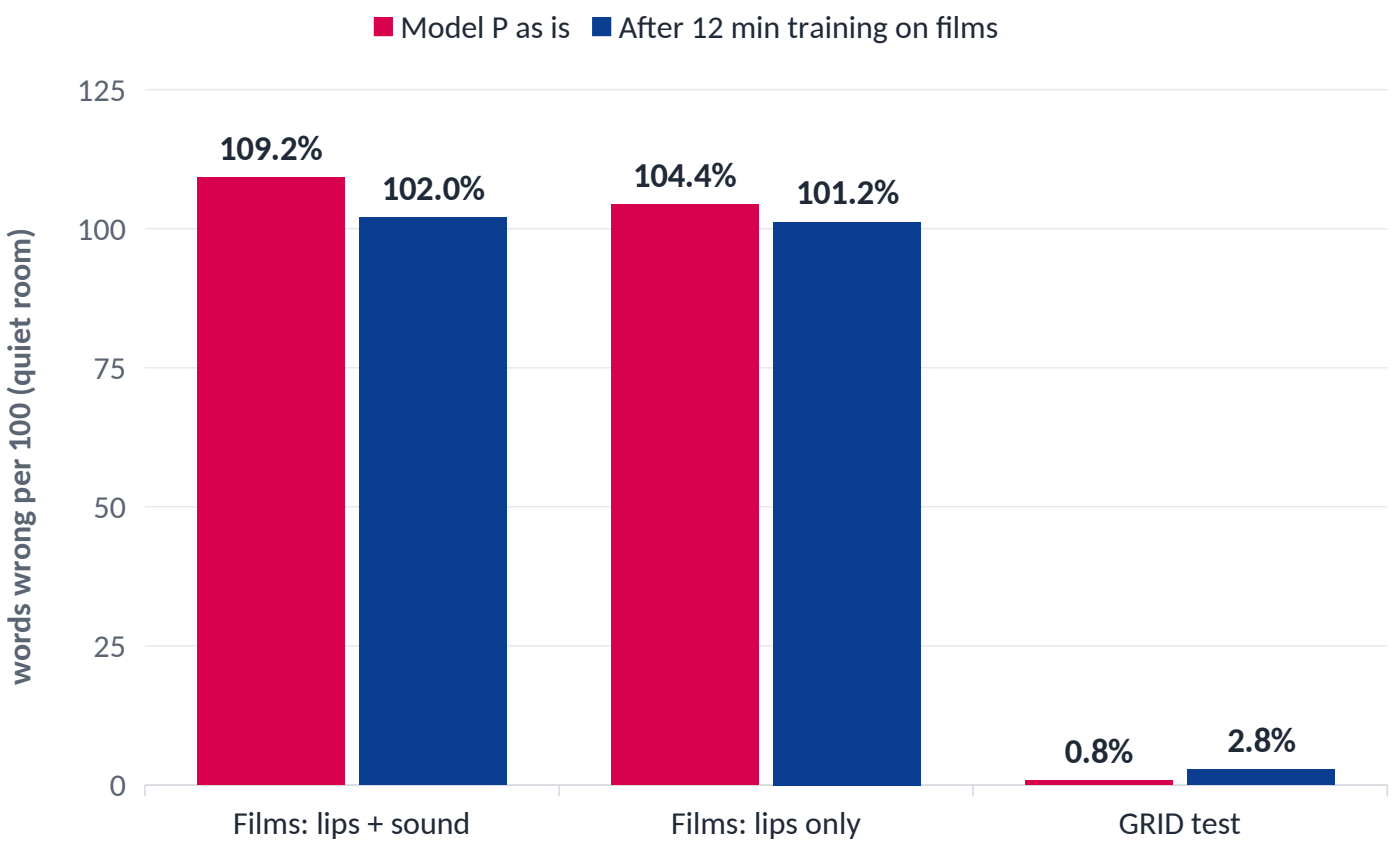
Training both senses from zero let the weak lip side add noise. Fixed by starting from the trained B1 and B2.

6 Crashes when a sense was missing

Blanking a whole sense made the maths divide by zero. The learned 'stand-in' signal fixed it.

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A first test outside GRID: film and TV clips



What we tried

305 clips from 13 films and series: 9.7 minutes, 651 different words. Tested on 3 films it never saw.

What we learned

Almost every film word is still wrong. Ten minutes is far too little; real systems train on hundreds of hours. Next: a proper dataset (LRS3 / LRW).

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What's next

✓ Just finished

Better sound features: ears-only errors on the tuning speakers dropped from 7.7% to 6.9%. To be merged.

First version of the caption app is in the code base.

M6 By 22 November

Retrain Model P with the new sound features.

Fine-tune settings and finish the full test report.

M8 By 29 November

Live captions from a webcam and microphone, powered by Model P.

Final testing and the demo.

Double-checked on the Alienware (12 Nov): all 9 trained models are saved, the 'best model' file is byte-for-byte the P4 model, and the test scores in this deck match the saved results.

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